Final Report on Olist Customer Satisfaction Analysis

NOTE: Except points 1, 2, 4, and 14, you can add/remove subpoints if necessary.

1. Name of the Project [1 mark]

Project Name	Predicting Customer Satisfaction for the Olist Brazilian E-Commerce Marketplace
Date	27 July 2026
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Executive Summary	Reviews arrive only after an order closes and are left by a minority of customers, so Olist cannot rely on them to save the order they describe. This project merges Olist's nine relational tables into a ~95,823-order dataset, engineers leakage-safe features, and trains a LightGBM classifier that predicts satisfaction (79%/21% class split) at the point of delivery. The tuned model reaches ROC-AUC 0.715 and, at a recall-optimized threshold, catches 72.5% of unsatisfied orders. SHAP confirms delivery delay, absolute delivery time, and seller history as the dominant drivers, pointing to a concrete first action: a live at-risk queue for in-transit orders trending late.

2. Problem Statement [1 mark]

Olist has no way to act on a bad experience until a review is already posted -- by then the order relationship is closed. This project frames satisfaction prediction as a binary classification problem (satisfied vs. unsatisfied) scored at the point of delivery, using delivery timing, seller history, and order attributes as predictors, so operations can intervene inside the window that still matters.

3. Project Goals and Objectives [1 mark]

Goal	Description	Criteria
Predict order-level satisfaction	Build a binary classifier that scores each delivered order as likely satisfied or unsatisfied before the review is left.	ROC-AUC >= 0.70 and recall on the unsatisfied class >= 70% on a held-out test set.
Identify the key drivers of dissatisfaction	Use SHAP/LIME to rank which order, delivery, and seller attributes most influence the prediction.	A ranked, business-readable driver list validated against the EDA (e.g., delivery delay confirmed as #1).
Enable proactive intervention	Translate model output into a dashboard and recommendation set operations can act on before an order is reviewed.	A dashboard, SWOT-framed narrative, and at least 3 concrete, measurable recommendations delivered.

4. Scope of the Project [1 mark]

In Scope	Delivered and reviewed Olist orders, Sep 2016-Aug 2018, all Brazilian states and product categories; data cleaning, EDA, leakage-safe modelling, explainability, and a business-facing dashboard/report/presentation.
Out of Scope	Real-time/streaming scoring, deployment to production infrastructure, review-text NLP, and any order outside the delivered+reviewed population (cancelled, in-transit, unreviewed).

5. Assumptions [1 mark]

Assumption ID	Assumption
A1	The Kaggle Olist extract is a representative, unmodified sample of the platform's real order history.
A2	review_score reflects genuine customer experience and is not subject to systematic manipulation or fraud.
A3	order_estimated_delivery_date is a fair proxy for the promise actually shown to the customer at checkout.
A4	Delivery timing and seller quality remain the dominant satisfaction drivers going forward (no major change in Olist's logistics network or seller base since 2018).

6. Constraints [1 mark]

Dependency ID	Description
C1	Single historical snapshot (2016-2018) -- no live data feed to validate the model on current orders.
C2	Class imbalance (79/21) caps achievable precision on the minority class at any reasonable recall level.
C3	No access to review text, customer service logs, or seller-side operational data that could add signal.
C4	Compute environment used for this submission is single-core, which bounded search-space size for hyperparameter tuning (documented in Notebook 3).

7. Dependencies [1 mark]

Dependency ID	Dependency	Impact
D1	Notebook 1 output (olist_clean.csv) feeding Notebook 2	Any schema change upstream breaks feature engineering and the leakage-safe split downstream.

D2	Notebook 2 output (olist_train/test_model.csv, baseline_metrics.csv) feeding Notebook 3	Model comparison and SHAP explainability cannot run without these exact files.
D3	Kaggle olistbr/brazilian-ecommerce dataset availability	Report and dashboard cannot be reproduced if the source dataset is altered or removed.
D4	Python libraries: pandas, scikit-learn, lightgbm, xgboost, shap, lime	Version drift in any of these can change exact metric values on re-run.

8. Risks and Mitigation [1 mark]

Risk ID	Risk Description	Mitigation Strategy
R1	Low precision on the unsatisfied class triggers too many unnecessary interventions.	Use the tuned probability threshold as a starting point and adjust to the business's actual cost of a false alarm vs. a missed at-risk order.
R2	Model trained on 2016-2018 data may not reflect current logistics/seller conditions.	Re-validate the model on a recent sample before any production use; retrain on a rolling window.
R3	Seller/category-level drivers could reflect confounded regional or seasonal effects, not causal seller quality.	Treat SHAP output as a prioritization signal for investigation, not a standalone quality verdict on a seller.
R4	Over-fitting during hyperparameter tuning given the constrained search budget.	Bias-variance check already run in Notebook 3 (train vs test AUC gap); re-check after any re-tuning.

9. Stakeholders (Stakeholder Analysis Matrix) [1 mark]

Stakeholder	Role	RACI	Interest and Influence
Olist Operations team	Acts on at-risk order flags	R (Responsible)	HIGH
Seller Management team	Owns seller onboarding/monitoring tiers	A (Accountable) for seller actions	HIGH
Data/Analytics team	Maintains and re-trains the model	R (Responsible)	MEDIUM
Executive sponsor	Approves budget for proactive-outreach program	I (Informed) / C (Consulted)	MEDIUM

10. Business Requirements (MoSCoW Prioritisation) [1 mark]

Priority	Requirement	Justification

MUST	Predict order-level satisfaction with recall >= 70% on the unsatisfied class.	Directly serves the core business goal of catching at-risk orders in time to act.
MUST	Leakage-safe train/test split with all statistics fit on training data only.	Without this, reported performance would not be trustworthy or reproducible in production.
SHOULD	Explain predictions with SHAP/LIME down to the individual order.	Operations needs a reason to act on a flag, not just a probability.
COULD	Extend to a live, streaming scoring pipeline.	Valuable but out of scope for this analysis; flagged for a future phase.

11. Evaluation Metrics [1 mark]

Metric	Target	How It Is Measured	Achieved
ROC-AUC	>= 0.70	sklearn roc_auc_score on the held-out test set	0.715 (tuned LightGBM)
Macro F1	> Stage-2 baseline (0.639)	sklearn f1_score(average='macro') on the test set	0.643
Recall (unsatisfied class)	>= 70% at chosen threshold	recall_score(pos_label=0) after threshold tuning	72.5% (tau=0.4)
Precision (unsatisfied class)	Reported alongside recall, no hard target	precision_score(pos_label=0) at the same threshold	29.7%

12. Sequence of Tasks Performed with Justification [1 mark]

(a) Merged the 9 raw Olist tables to one row per order_id, aggregating multi-item/multi-payment orders before joining -- otherwise the merge multiplies rows and corrupts the target distribution.

(b) Restricted to delivered + reviewed orders only, since both a delivery event and a review are required to define the target; dropped the single order missing a payment record as a genuine data-quality gap.

(c) Performed the train/test split BEFORE deriving any statistic (imputation medians, frequency encodings, out-of-fold seller_avg_score) so nothing from the test set could leak into training.

(d) Compared 4 models by 5-fold CV ROC-AUC, tuned the winner (LightGBM) with RandomizedSearchCV, reduced to 16 features by cumulative importance, then explained it with SHAP and LIME before exporting predictions.

13. Model Selection Summary [1 mark]

Model	Accuracy	Macro F1	ROC-AUC	Recall (unsatisfied)
Logistic Regression (baseline)	74.2%	0.639	0.709	49.4%
Random Forest	76.3%	0.652	0.711	47.0%
XGBoost	73.4%	0.636	0.703	51.1%
LightGBM (tuned, final)	74.1%	0.643	0.715	51.5%

14. Analytical Findings [1 mark]

(a) Key drivers of dissatisfaction - delivery_delay_days, actual_delivery_days, and seller_avg_score dominate the SHAP ranking; on-time orders are satisfied 82.7% of the time vs. 26.7% for late orders, and satisfaction is lowest in North/Northeast states farther from the main Southeast logistics hubs.

(b) Limitations - Precision on the unsatisfied class remains modest (30-45%) even after tuning; the model improves on the Stage-2 baseline only incrementally (ROC-AUC +0.0055), suggesting most of the learnable signal in this feature set is already captured and further gains likely need richer data (review text, seller ops detail).

15. Business Recommendations [1 mark]

(a) Build a live at-risk queue for in-transit orders whose tracking already shows they will miss the estimated delivery date, and trigger proactive outreach before the review is left.

(b) Route new or low-volume sellers who share risk factors with historically low-scoring sellers into a stricter onboarding and monitoring tier.

(c) Prioritize a shipping-cost and packaging review for office_furniture and home_confort, the two lowest-satisfaction categories with meaningful order volume.

(d) Build fulfilment buffer capacity ahead of known seasonal demand spikes (Black Friday, the recurring Q1 dip) to prevent the satisfaction drops seen in Nov 2017 and Feb-Mar 2018.

16. Glossary [1 mark]

Term	Definition
satisfied (target)	Binary label: 1 if review_score >= 4 (4-5 stars), else 0 (1-3 stars).
delivery_delay_days	order_delivered_customer_date minus order_estimated_delivery_date, in days; positive means delivered late.
seller_avg_score	A seller's historical average satisfaction rate, computed out-of-fold on training data only to avoid leakage.
SHAP / LIME	Model-explainability techniques that attribute a prediction to individual feature contributions, globally (SHAP summary) or for one order (SHAP waterfall / LIME).